

Ashish Raj

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PROFESSIONAL SUMMARY

Backend-focused Software engineer with production experience building scalable systems and REST APIs. Proficient in system design, cloud computing, caching strategies, and Docker containerization. Hands-on with WebSockets, RESTful APIs, remote servers, and deploying services on cloud infrastructure. Currently expanding skills in Golang and Kubernetes. Shipped a production delivery app on both app stores with 500+ active users, handling real-time order processing, JWT authentication, caching, and Firebase integrations.

TECHNICAL SKILLS

Languages: Python, Java, Node.js, C, SQL, Golang (actively learning)

Backend and Distributed Systems: FastAPI, Express.js, REST APIs, Microservices, WebSockets, Multithreading, Memory Management

System Design: Load Balancing, Horizontal Scaling, Caching Strategies, Event-Driven Architecture

Caching and Databases: Redis, PostgreSQL, Supabase, Database Indexing, Query Optimization

Cloud Computing and DevOps: AWS (EC2, S3), Docker, Kubernetes (actively learning), CI/CD Pipelines (GitHub Actions), VPS Deployment

Security and Auth: JWT Authentication, OTP Systems, Token-Based Authorization, Role-Based Access Control (RBAC)

Operating Systems: Linux (Ubuntu, Debian), Process Management, Memory Management, File Systems, Shell Scripting

Tools: Git, Postman, Docker Compose

PROFESSIONAL EXPERIENCE

Backend Intern - Propell Action Private Limited

June 2026 - August 2026

Hair & Skin AI App (Consumer Health-Tech)

Remote

- Migrated the product off a low-code AI app-builder to a standalone Vite, React, and Supabase codebase, and built a Flutter mobile client so the web and mobile apps could share the same backend services.
- Designed and shipped a dynamic role-based access control (RBAC) system with configurable roles and permissions, an admin console, and admin-invite workflows, replacing static role checks across the platform.
- Strengthened security and compliance by adding DPDP/GDPR consent and data export/deletion flows, security headers, EXIF metadata stripping on image uploads, and rate limiting across serverless functions handling OTP, AI analysis, and subscription activation.

SDE Intern - ValueKare Technologies Private Limited

June 2025 - August 2025

Fixed Asset Management System

Jaipur

- Built a comprehensive fixed asset management system enabling organizations to track assets, manage transfers between departments, handle auditing workflows, process maintenance requests, manage user registration and authentication, and perform bulk asset uploads via Excel.
- Implemented role-based access control with granular permissions across 4 tiers (Admin, HOD, Auditor, Operator), containerized services using Docker Compose, and deployed on AWS with a CI/CD pipeline via GitHub Actions, reducing release cycles by 60 percent.

Software Development Intern - iGURUS Consultancy Services

June 2025 - August 2025

Backend Services and Real-Time Analytics

Remote

- Built a financial comparison platform using FastAPI and PostgreSQL enabling users to compare credit cards based on their spending habits, receive personalized recommendations, and discover beneficial government and banking schemes through web-scraped data, Excel uploads, and structured database design.
- Improved backend performance by 35 percent through database indexing, Redis caching, and optimized SQL queries while designing high-availability architecture with connection pooling for concurrent request handling.

PROJECTS

Localised Food and Grocery Delivery Platform (Currently active)

January 2024 - Present

Flutter, Node.js, PostgreSQL, Redis

- Built and launched a full-stack food and grocery delivery platform serving 500+ active users with real-time order processing, JWT authentication, OTP login via Firebase, and push notifications and published it on both the Google Play Store and Apple App Store.
- Architected the backend on a Linux VPS with PostgreSQL database design; improved API performance by 40 percent using Redis caching and query optimization; and managed end-to-end deployment pipelines, including build configuration and signing certificates.

HACRX: AI-Powered Legal Document Retrieval System

July 2025 - August 2025

Python, GPT-4, Pinecone, OpenAI Embeddings

- Built an LLM-driven semantic search system processing 10,000+ legal documents with sub-100 ms retrieval using the Pinecone vector database and OpenAI text embeddings.
- Engineered the end-to-end data pipeline, including preprocessing, embedding generation, chunking strategy, and relevance ranking, with an 80 percent reduction in manual search time.

Automated Warehouse Climate Control - Industrial IoT

ESP32, Python, LSTM, Reinforcement Learning, REST APIs

- Developed an Industrial IoT system using ESP32 sensors transmitting real-time environmental data via RESTful APIs for warehouse temperature and humidity monitoring.
- Implemented LSTM-based prediction and reinforcement learning control for automated climate optimization, applying edge computing for low-latency decisions.
- Built a real-time monitoring dashboard with automated alerts covering 5+ warehouse zones, reducing spoilage incidents.

EDUCATION

B.Tech. in Computer Science and Engineering

August 2023 - Present

National Institute of Technology Sikkim | CGPA: 7.02 (Till 6th Semester)

Senior Secondary (XII), Science - Shivam Convent School

2022

Percentage: 83 percent

Secondary (X)—Jawahar Navodaya Vidyalaya Saran

2020

Percentage: 86.7 percent

CERTIFICATIONS

- Flutter and Dart - The Complete Guide
- Mastering Data Structures and Algorithms using C

LEADERSHIP AND ACTIVITIES

Backend Lead - Food delivery application

January 2024 - Present

- Led technical architecture decisions and deployment strategy while scaling the platform to serve real users in production.

Core Member - The Regnant Ink (Literary Club, NIT Sikkim)

April 2024 - Present

- Organized 8+ events including workshops and competitions, coordinating across teams to engage 50+ participants.